out by *substituting variables*, just as with the bilinear transform. We start by writing the transfer function of the unity cutoff filter, and then replace each z^{-1} with the following:

EQUATION 33-12
Low-pass to low-pass transform. This is a method of changing the cutoff frequency of low-pass filters. The original filter has a cutoff frequency of unity, while the new filter has a cutoff frequency of *W*, in the range of 0 to π.

$$z^{-1} \rightarrow \frac{z^{-1} - k}{1 - kz^{-1}}$$

where:

$$k = \frac{\sin(1/2 - W/2)}{\sin(1/2 + W/2)}$$

This provides the transfer function of the filter with the new cutoff frequency. The following design equations result from applying this substitution to the biquad, i.e., no more than two poles and two zeros:

EQUATION 33-13
Low-pass to low-pass conversion. The recursion coefficients of the filter with unity cutoff are shown in italics. The coefficients of the low-pass filter with a cutoff frequency of *W* are in roman.

$$\mathrm{a}_0 = (a_0 - a_1 k + a_2 k^2)/D$$
$$\mathrm{a}_1 = (-2a_0 k + a_1 + a_1 k^2 - 2a_2 k)/D$$
$$\mathrm{a}_2 = (a_0 k^2 - a_1 k + a_2)/D$$
$$b_1 = (2k + b_1 + b_1 k^2 - 2b_2 k)/D$$
$$b_2 = (-k^2 - b_1 k + b_2)/D$$

where:

$$D = 1 + b_1 k - b_2 k^2$$
$$k = \frac{\sin(1/2 - W/2)}{\sin(1/2 + W/2)}$$

Low-pass to High-pass Frequency Change

The above transform can be modified to change the response of the system from low-pass to high-pass while simultaneously changing the cutoff frequency. This is accomplished by using a **low-pass to high-pass transform**, via the substitution:

EQUATION 33-14
Low-pass to high-pass transform. This substitution changes a low-pass filter into a high-pass filter. The cutoff frequency of the low-pass filter is *one*, while the cutoff frequency of the high-pass filter is *W*.

$$z^{-1} \rightarrow \frac{-z^{-1} - k}{1 + kz^{-1}}$$

where:

$$k = -\frac{\cos(W/2 + 1/2)}{\cos(W/2 - 1/2)}$$

As before, this can be reduced to design equations for changing the coefficients of a biquad stage. As it turns out, the equations are identical